CL 299 presentation 5

LNP Manufacturing and Process Simulation

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Energy Balance

Equipment	Typical Lab Power Range
Syringe pump	5–20 W
Magnetic stirrer	5–15 W
Peristaltic pump	20–60 W
TFF circulation pump	80–120 W

Ptotal = **200 W**
Energy consumption of the system is calculated using the equation:

$$E = P \times t$$

E = 0.22 kWh

What is in the Code?

- A Python-based simulator that takes feed compositions and flow rates as input and automatically calculates every stream in the process — including TFF splits and ethanol removal.

What You Input in the Code

- **Molar Ratios — Organic Phase**
  Ionizable Lipid
  Helper Lipid Cholesterol
  PEG Lipid
  Cholesterol
  Ratios converted to wt% using MW of each component

- **Molar Ratios — Aqueous Phase**
  mRNA
  Buffer
  Absolute concentrations set separately as mg/mL inputs

- **Feed Flow Rates**
  Aqueous feed flow (mL/min)
  Organic feed flow (mL/min)
  Changing these two values changes every downstream flow automatically

- **Feed Concentrations**
  mRNA conc in aqueous feed (mg/mL)
  Buffer salt conc (mg/mL)
  Total lipid conc in organic feed (mg/mL)

- **Process Parameters**
  Encapsulation efficiency (fraction, e.g. 0.90)
  Target ethanol vol% (e.g. 8.0%)
  TFF Volume Reduction Ratio — VRR
  Diavolumes N (for $C = C_0 \cdot e^{-N}$)
  LNP retention at membrane
  Salt sieving coefficient
  Sterile filter dead volume (mL)
  Process parameters are fixed equipment settings — not changed per run

Key Equations Used in the Model

Mole Fraction → Mass Fraction
$wf_i = (x_i \times MW_i) / \Sigma(x_j \times MW_j)$

Diafiltration Ethanol Removal ($C = C_0 \cdot e^{-N}$)
$C\_ethanol = C_0 \times \exp(-N)$
Standard diafiltration washout equation. N = number of diavolumes. N=5 → 99.3% removal. Used in FDA submissions and industry protocols.

Dilution Buffer Calculation
$V\_dil = V\_EtOH / (target\% / 100) - V\_S12$
Back-calculates dilution buffer volume required to reduce ethanol from post-mixer level to target vol%. Adjusts automatically for any FRR.

Buffer Salt Removal with Sieving Coefficient
$C\_salt = C_0 \times \exp(-\sigma \times N)$
Extension of ethanol model. σ (sieving coefficient) = fraction passing per diavolume. For 100 kDa membrane + small salts, $\sigma \approx 1.0$.

TFF Volume Reduction Ratio
$VRR = V\_S13 / V\_retentate$

Sample Output (aq = 9 mL/min, org = 3 mL/min)

Stream	Flow (mL/min)	Ethanol (vol%)	LNP Mass Flow (mg/min)
Aqueous Feed	9.00	—	2.70 (mRNA)
Organic Feed	3.00	98.7%	30.00 (lipids)
Crude LNP (S12)	12.00	24.7%	32.70
Dilution Buffer	25.02	—	—
Diluted LNP (S13)	37.02	8.0%	32.70
DF Buffer → TFF	37.02	—	—
TFF Permeate	66.64	~99%→0	1.65 (lost)
TFF Retentate	7.40	0.2%	29.28
Final Product	7.40	0.2%	29.28

Balance Error
0%

Encapsulated mRNA Recovery
89.1%

Ethanol Removed (N=5 diavolumes)
99.33%

What the Simulator Outputs

- **Feed Streams**
  Flow rate (mL/min)
  Concentration of each component (mg/mL)
  Mass flow of each component (mg/min)

- **Final Product Stream**
  Final product flow rate (mL/min)
  Concentration of all components in product
  Mass flow of each component in final product
  Encapsulated mRNA mass flow and recovery %

- **Calculated Process Flows**
  Dilution buffer added (auto-calculated from target ethanol %)
  Ethanol vol% at each stage
  DF buffer flow (auto-scaled from diavolumes × retentate)
  TFF permeate flow (auto-calculated from VRR)

- **Balance Verification**
  Total mass IN vs total mass OUT
  Volumetric flow accounting (IN = OUT)
  Balance error (%)
  Sterile filter dead volume noted separately

PAT

PAT classification - INLINE

Now we have classification of PAT techniques used in LNP manufacturing.

- Dynamic light scattering(DLS)
  measure particle size and distribution & PDI (<0.2).

- Zeta potential analysis
  measure surface charge that indicates the stability of suspension.

- Optical turbidity measurement(turbidity probes)
  it detects incorrect lipid mixing
  aggregation and wrong molecule formation.

- PVM
  capturing image while LNPs are forming

- NIR
  monitor composition and concentration of components in real time without sampling.

- UV–Vis Spectroscopy
  it monitors Concentration of lipids, Encapsulated drug / RNA, Particle formation kinetics and Impurities.
- pH sensor
  at the buffer mixing for correct pH.
- flow meter
  measure flow rate of liquid streams.
- temp prob
- conductivity sensor
  buffer dilution verification.
- Pressure sensor
  in TFF filtration.
- Raman spectroscopy

Online

- High-Performance Liquid Chromatography (HPLC)
  On-line possible via auto-samplers for drug loading, encapsulation efficiency, or degradation monitoring.

- Raman spectroscopy
  used as a downstream analytical tool to measure how much mRNA is successfully encapsulated inside lipid nanoparticles.

- Charged aerosol detectors (CAD)
  Used with chromatography to detect and quantify lipid components in LNP formulations that do not absorb UV light

- Refractive index (RI)
  Measures changes in refractive index to detect lipids and polymers in LNP samples during chromatographic analysis.

- ultraviolet(UV) diode array
  Detects UV-absorbing molecules such as RNA or drugs in LNP formulations while simultaneously recording their absorption spectra

- LC-MS
  lipid identification and impurity detection

Atline

- Raman spectroscopy
- Zeta potential analysis
- DLS

- NMR
  lipid structure and molecular interactions
- Cryo-TEM
  Visualize morphology and internal structure of LNPs
  confirm nanoparticle formation and observe structure of lipid core.

Resources:

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THANK YOU!!!